



Lightning - v3

Domain Portfolio: Observations | Domain: Historical | Usage Classification: Standard

Geography: Global

Attribution Required: N/A

Attribution Requirements: N/A

Overview

The Lightning API provides the ability to lookup lightning strikes that happened for the given period of time at the specified location (within 10 miles or 17 kilometers). This API supports search by: Geocode (Latitude and Longitude), Postal Key, IATA Code, ICAO Code and Place ID. This API provides recent lightning strikes data for a 15 minute or 60 minute period from the request.

HTTP Headers and Data Lifetime - Caching and Expiration

For details on appropriate header values as well as caching and expiration definitions, please see [The Weather Company Data | API Common Usage Guide](#).

Note: This product offers different endpoints for mobile vs. desktop usage due to differing contractual requirements. Due to related contractual limitations, 59 minutes of data will be returned for the 60minute/mobile endpoint and 55 minutes of data will be returned for the 60minute/desktop endpoint. (See Appendix: Endpoint Diagrams).

Note: The lightning/**insight**/ endpoint is unique in that it provides only the SINGLE closest lightning strike that happened for the given period of time at the specified location (within 10 miles / 17 kilometers).

Atomic Endpoints	Aggregate Product Name
v3/wx/lightning/15minute/mobile	v3-wx-lightning-15minute-mobile
v3/wx/lightning/insight/15minute/mobile	v3-wx-lightning-insight-15minute-mobile
v3/wx/lightning/15minute/desktop	v3-wx-lightning-15minute-desktop
v3/wx/lightning/60minute/mobile	v3-wx-lightning-60minute-mobile
v3/wx/lightning/60minute/desktop	v3-wx-lightning-60minute-desktop

URL Construction

Request by Geocode: Required Parameters: <code>format</code> , <code>units</code> , <code>geocode</code> , <code>apiKey</code> Optional Parameters: <code>strikeType</code> <code>https://api.weather.com/v3/wx/lightning/15minute/mobile?geocode=<geocode>&format=<format>&units=<units>&apiKey=yourApiKey</code>
<code>https://api.weather.com/v3/wx/lightning/15minute/mobile?geocode=33.74,-84.39&format=json&units=e&apiKey=yourApiKey</code> Note that there are limitations on the latitudes supported. Invalid latitudes above -85,85 will return a 400 status code.
Request by IATA Code: Required Parameters: <code>format</code> , <code>units</code> , <code>iataCode</code> , <code>apiKey</code> Optional Parameters: <code>strikeType</code>

https://api.weather.com/v3/wx/lightning/15minute/mobile?iataCode=<iataCode>&format=<format>&units=<units>&apiKey=yourApiKey
https://api.weather.com/v3/wx/lightning/15minute/mobile?iataCode=BOS&format=json&units=e&apiKey=yourApiKey
Request by ICAO Code: Required Parameters: format, units, icaoCode, apiKey Optional Parameters: strikeType https://api.weather.com/v3/wx/lightning/15minute/mobile?icaoCode=<icaoCode>&format=<format>&units=<units>&apiKey=yourApiKey
https://api.weather.com/v3/wx/lightning/15minute/mobile?icaoCode=KBOS&format=json&units=e&apiKey=yourApiKey
Request by Postal Key: Required Parameters: format, units, postalKey, apiKey Optional Parameters: strikeType https://api.weather.com/v3/wx/lightning/15minute/mobile?postalKey=<postalKey>&format=<format>&units=<units>&apiKey=yourApiKey
https://api.weather.com/v3/wx/lightning/15minute/mobile?postalKey=01810:US&format=json&units=e&apiKey=yourApiKey
Request by Place ID: Required Parameters: format, units, placeid, apiKey Optional Parameters: strikeType https://api.weather.com/v3/wx/lightning/15minute/mobile?placeid=<placeid>&format=<format>&units=<units>&apiKey=yourApiKey
https://api.weather.com/v3/wx/lightning/15minute/mobile?placeid=327145917e06d09373dd2760425a88622a62d248fd97550eb4883737d8d1173b&format=json&units=e&apiKey=yourApiKey

Valid Parameter Definitions

Parameter	Description	Example
format	The format in which the response is to be returned.	json
units	For units = e, the output fields will be in miles. For units = m, the output fields will be in kilometers.	e
geocode	The geocode of the location for which lightning strikes is to be returned. The geocode is in the format latitude,longitude. The latitude and longitude is a double value having precision of not more than 6 digits.	42.375,-71.039
iataCode	The International Air Transport Association (IATA) airport code for which lightning strikes is to be returned.	BOS
icaoCode	The International Civil Aviation Organization (ICAO) airport code for which lightning strikes is to be returned.	KBOS
postalKey	Postal Key is a composite location identifier key of postal code:country code.	01810:US
placeid	A unique place identifier.	779593cd7a42a5d9f680b0f136acc297dfddbc942e3c1b5a550476dba0bcd76d
strikeType	strikeTypeCode of cloud = cloud to cloud strike type strikeTypeCode of ground = cloud to ground strike type strikeTypeCode of all = both cloud to cloud AND cloud to ground strike types Note - not valid for ‘insight’ endpoint	cloud

Data Elements & Definitions

Note: Field names are sorted alphabetically in the table below for presentation purposes. The table below does not represent the sort order of the API response.

Field Name	Description	Type	Range	Sample	Nulls Allowed
"lightning"					
distance	Distance from the location specified to all lightning strikes within the radius during the specified time period	[decimal]	0 to 10 miles, 0 to 17 kilometers	1.21168	N
latitude	Latitude of strike location up to five (5) decimals	[decimal]	-90.0 to 90.0	-1.79979	N
longitude	Longitude of strike location up to five (5) decimals	[decimal]	-180.0 to 180.0	-44.54565	N
validTimeUtc	Lightning strike time detected by the network	[ISO]		2019-06-17T06:57:00	N

v3/wx/lightning/15minute/mobile - JSON Sample

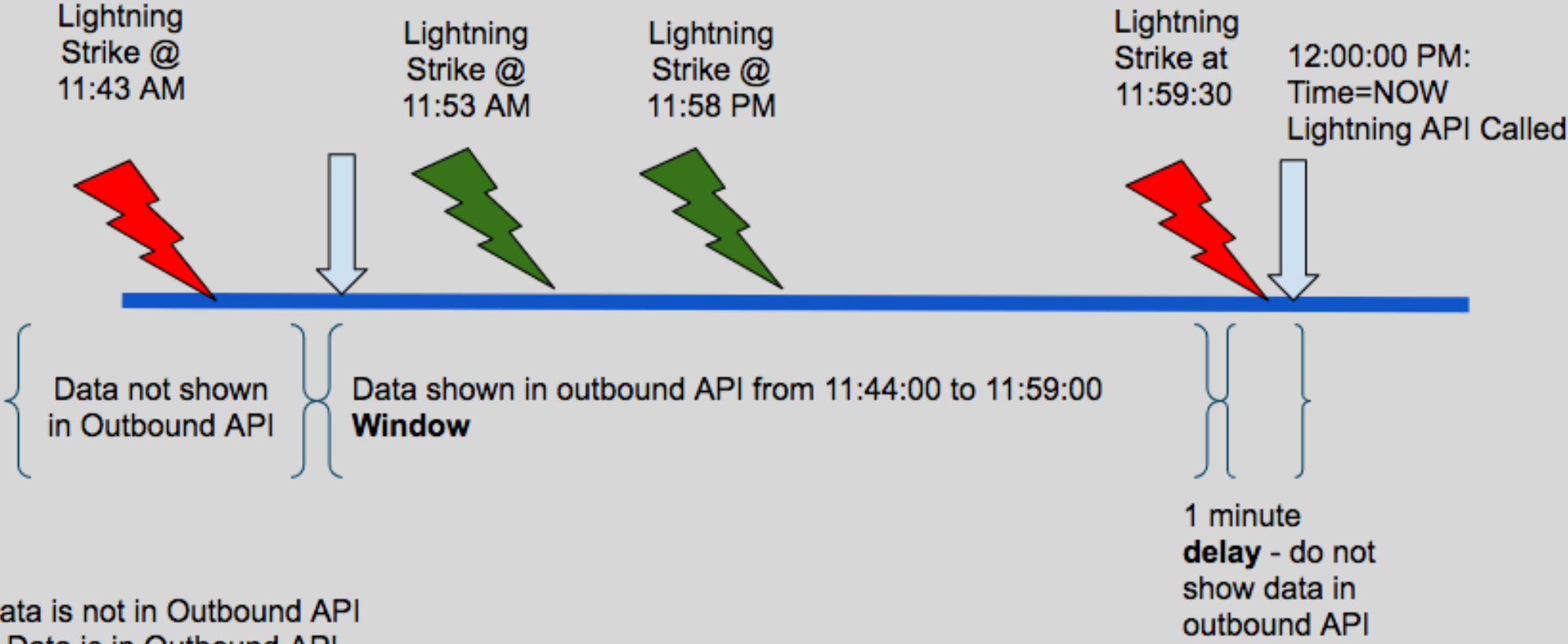
```
{
  "distance": [
    0.2957,
    4.9248,
    7.9516,
    9.9674
  ],
  "latitude": [
    35.98173,
    36.12602,
    35.86129,
    35.86122
  ],
  "longitude": [
    -96.77026,
    -96.80117,
```

```
-96.85778,  
-96.85819  
],  
"validTimeUtc": [  
  "2019-08-23T12:59:00",  
  "2019-08-23T13:02:53",  
  "2019-08-23T13:08:07",  
  "2019-08-23T13:08:07"  
]  
}  
}
```

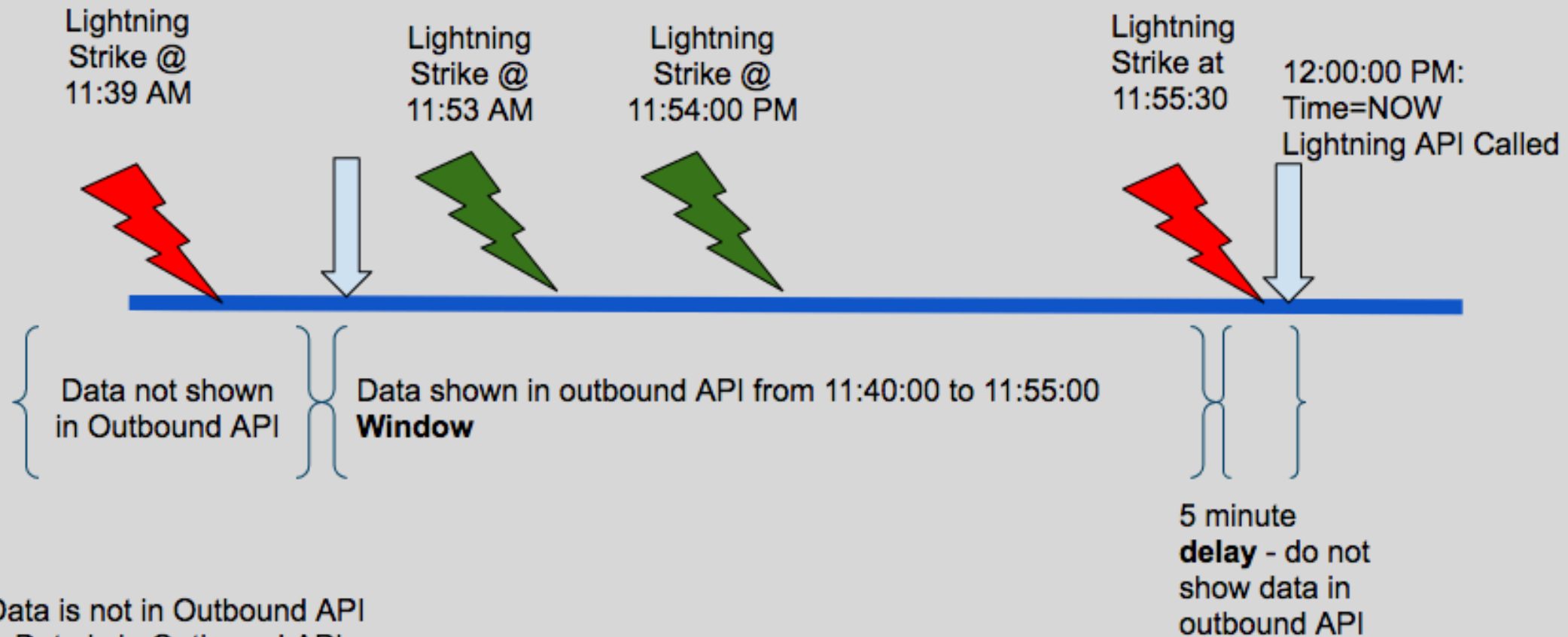
[v3/wx/lightning/insight/15minute/mobile](#) - JSON Sample

```
{  
  "distance": [  
    0.2957  
  ],  
  "latitude": [  
    35.98173  
  ],  
  "longitude": [  
    -96.77026  
  ],  
  "validTimeUtc": [  
    "2019-08-23T12:59:00"  
  ]  
}
```

15minute/mobile

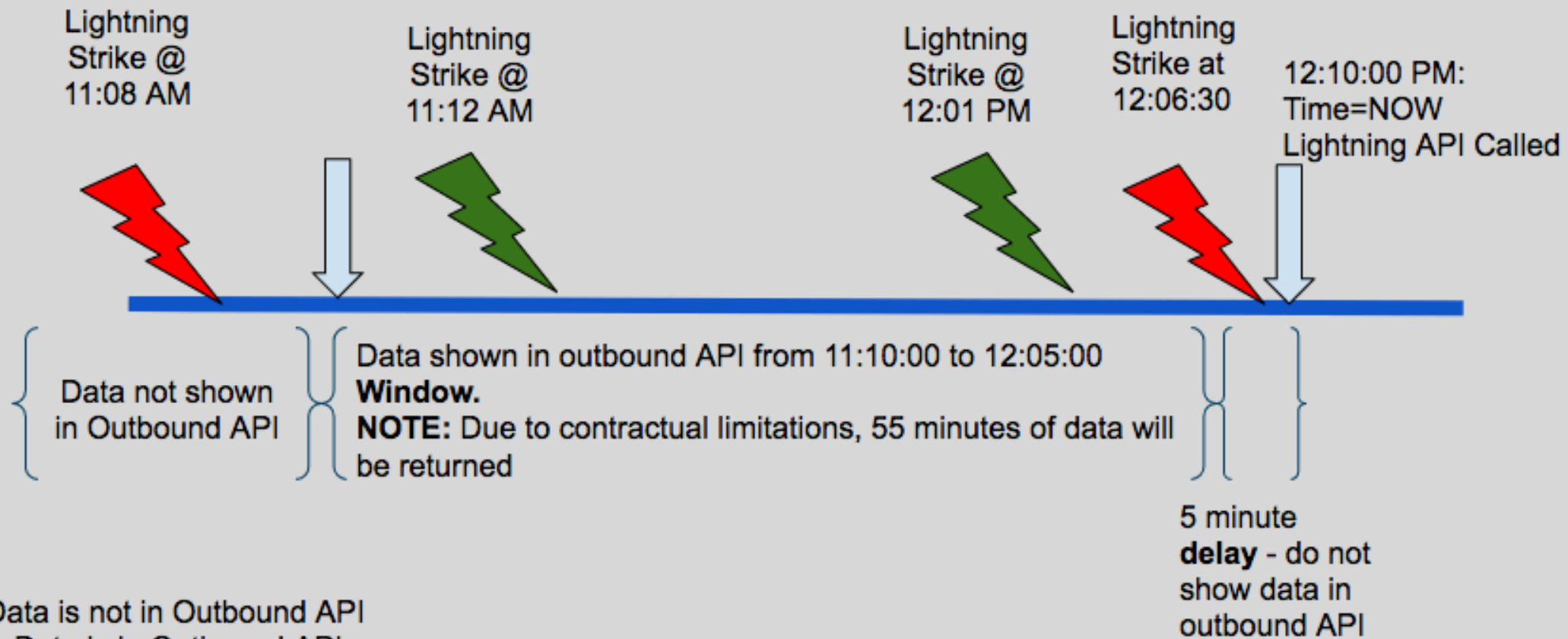


15minute/desktop



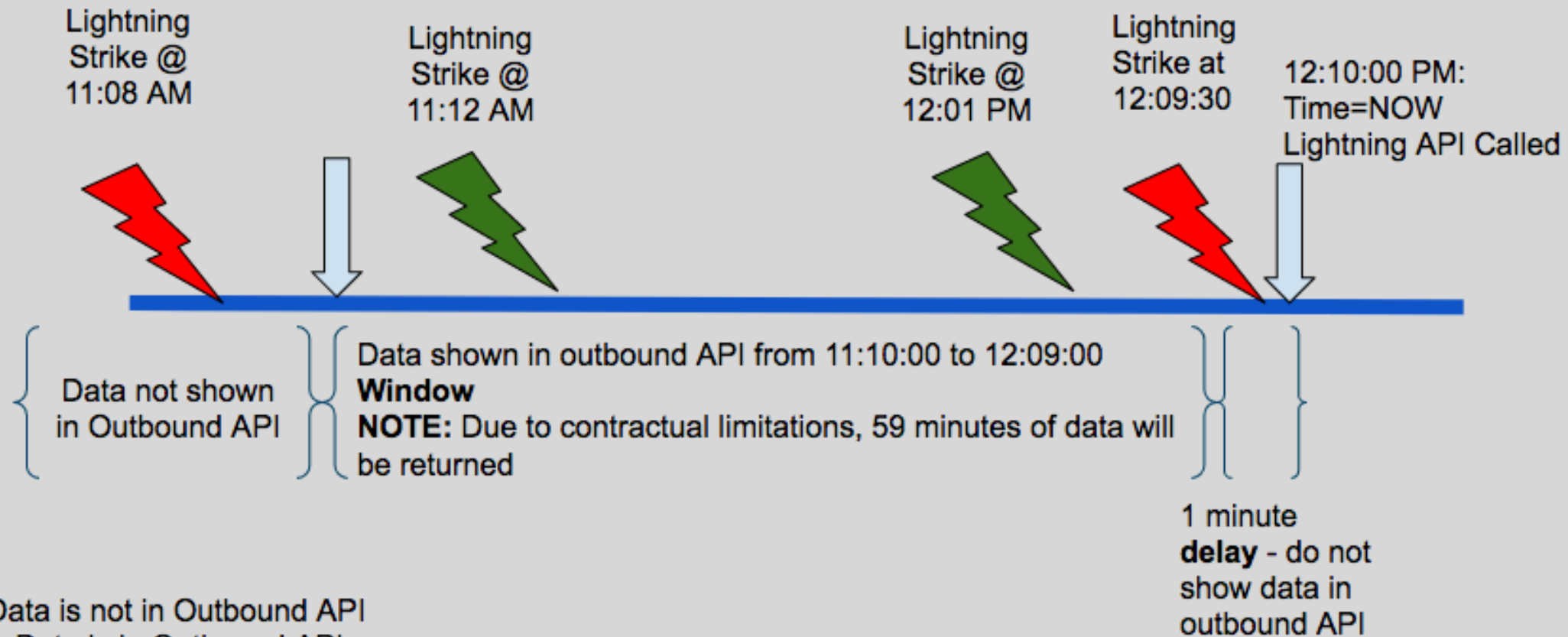
Red = Data is not in Outbound API
Green = Data is in Outbound API

60minute/desktop



Red = Data is not in Outbound API
Green = Data is in Outbound API

60minute/mobile



Red = Data is not in Outbound API
Green = Data is in Outbound API